B1 LoRMA+ statute classifier: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Test whether a multiplicative low-rank adaptation changes the inherited B1 representation more effectively than additive LoRA. Only the adaptation mechanism changes relative to B1 LoRA; that makes the pair useful for testing parameter-efficient adaptation rather than changing the task pipeline all at once.

Architecture and mechanism

LoRMA+ wraps a linear output with trainable low-rank multiplicative factors while preserving the pretrained Phase 1 weights. The same multi-label calibration and evaluation route is retained for a meaningful comparison.

Measured result

Scored submission	1016282
Codabench macro-F1	0.2542
Archived local metric	0.2517 official-dev macro-F1

The observed Codabench score, 0.2542, is slightly below LoRA's 0.2554. The difference is small and cannot establish a general ranking of the methods.

Limitations and next check

A low-rank output correction cannot recover evidence that the window selection never passes to the encoder. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase5_b1_train.py, phase5_lorma.py, phase5_core.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.